

Quantum Computing

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1 Abstract

This paper introduces the methods of Quantum Computing and quantum teleportation. It serves as an introduction to those who are curious. It is intended for readers with a background in Physics, familiar with Quantum Mechanics but with no previous exposure to quantum computing.

2 Introduction: The need for Quantum Computing

Present day technology is based on the silicon microprocessor chip. Silicon technology developed rapidly, with Gordon Moore (co-founder of Intel Corporation) proclaiming that the number of transistors on a chip would develop every 18-24 months. Now known as Moore's law, the prediction of exponential growth predicted has held true for a long period of time.

However, this cannot hold true for ever as the decreasing size of transistors introduces more complications. On inspection, more complexity arrives at the quantum scale, as the transistor approaches the size of the de-Broglie wavelength of the electrons that carry the signals. Thus, Moore's law must eventually halt to be true, imposing an upper bound on computational power using classical transistors.

2.0.1 Breif detour into Algorithmic Complexity

Computer algorithms are judged in efficiency by how the number of operations required for an algorithm to run complete., varies with respect to the size of the input. This rating must be done without regards to the power of the hardware the algorithm is running on, and thus a form of asymptotic analysis is used where the specifics of the hardware are ignored.

If the number of operations increases are a polynomial power, (i.e. proportional to N^k where N is the size of input and k is some integer) then the problem is said to belong to the **polynomial complexity class**, abbreviated to **P**. If, on the other hand, the number of operation increases faster than a polynomial function, then the problem belongs to the **non-polynomial complexity class** (**NP**).

Classical computers adequately deal with **P** problems, but not **NP** problems due to how they scale with a small change in the size of the input.

2.0.2 Enter Quantum Computation

The idea of running computers using the laws of Quantum Mechanics was initially proposed by Richard Feynman[1]. He made the claim that simulating quantum systems with conventional computing methods was difficult as the size of the system increased.

Quantum Computing relies on the fundamental principles of quantum mechanical behaviour, namely that of superposition and entanglement. Within classical computational methods, information is stored on bits, which can be in one of two states i.e. 0 or 1. Quantum computers store information on quantum bits – or qubits. A qubit is a two-state quantum system, which can be in a superposition of both states with respective complex amplitudes. That is the fundamental idea behind quantum computation.

A common misconception is that the superposition makes all processes faster. However, this is not the case. Quantum computing is powerful as it is able to simultaneously deal with a large number of variables. This is due to how a collection of qubits store data. For example, consider a collection of 2 classical bits which can be in one of the four configurations available (00, 01, 10 and 11). Qubits can be in a superposition of those states, that is the qubit can be in a state (via Dirac notation) $|\psi\rangle$ where

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

where the co-efficients of the states are complex. A collection of 4 bit or qubits, have 2^4 available configurations. Classical bits can only hold one of those 16 values at once, whereas qubits are able to be a superposition of the different available configuration state. Thus, to double the amount of computational power, classically this requires doubling the number of bits available. In contrast, through the use of quantum computing and qubits, only an additional qubit is required to achieve the same result. Using quantum logic gates, we can perform operations on these qubits, which return another superposition as an output.

The usefulness of this method of computing is best shown through a simple example, prime factorization. Given a integer N which is a product of two primes $p * q$, find p and q . The classical brute force method would be to check all the odd integers up to $\sqrt[3]{N}$, one by one. Through quantum computing, you are able to check multiple factors at the same time. For small numbers, this method is not necessarily faster. Using bits, the amount of time required to find p or q scales exponentially with the number of digits in N . Through quantum computational methods via Shor's Algorithm [2], the factors can be found in polynomial time, rather than exponential time. Shor's algorithm uses the quantum fourier transform to find the periodicity of a set of integer numbers.

The problem has been converted from finding the factors to one which requires finding the period of a function.

Mathematics is but one of the numerous applications of quantum computing. Quantum computation maybe applicable to any system dealing with a large number of variables. Currently, methods are being proposed to further progress to be applied in machine learning, quantum chemistry simulations which help with the manufacturing of pharmaceuticals. Quantum computational methods help in quantum chemistry by computing the large number of electron-electron interaction between atoms. Classically, this requires an intensive amount of computational power.

2.1 Breif recap of Quantum Mechanics

In the two-state system, the two different states are referred to as the basis states, as any other configuration can be de-composed into a superposition of these two basis states. When a qubit is measured, the result can only be of the basis states. Due to the nature of superposition, this measurement is probabilistic. When taking a measurement of a generic state with respect to a basis state, the measurement returns that basis state with a certain probability that is dependent on the weight of the basis state in the generic state. The probabilities are acquired from the norm of the complex amplitudes of each basis state. How the states along with the complex co-efficients behave (and thus the probabilities) are governed by the postulates of Quantum Mechanics.

2.2 Manipulating qubits using quantum gates

Within an algorithm manipulating qubits, the operations are performed by quantum logic gates. A powerful property that quantum gates and their operations posses is that they are reversible. This property follows from laws of quantum mechanics. Quantum gates can be represented by $2^n \times 2^n$ unitary matrices, where n is the number of qubits being acted upon. Unitary matrices preserve the norm of the states, that is to say that the sum of the probabilities remains 1 after an operation. This follows from a postulate of quantum mechanics which states that *the evolution of a closed quantum system is described by a unitary transformation*. Given the intial state $|\psi\rangle$ with $\langle\psi|\psi\rangle = 1$ and a unitary operation U, if $U|\psi\rangle = |\phi\rangle$, then $\langle\phi|\phi\rangle = 1$

For any quantum system, there is a unitary operator $U(t, t_0)$ such that for any state $|\Psi, t_0\rangle$ of the system at time t_0 the state at time t is obtained as:

$$|\Psi, t\rangle = U(t, t_0) |\Psi, t_0\rangle, \forall t, t_0$$

A unitary transformation U satisfies the condition $U^\dagger U = 1$, which implies $U^{-1} = U^\dagger$, and thus they are always reversible. Within quantum computation this can be thought of as a preservation of information from state to state, which is important for quantum states to remain measurable at the end of computation.

2.2.1 Density Matrix

The density matrix is a matrix which describes a quantum system composed of several states, in contrast to a pure state (a single state vector $|\Psi\rangle$). The density matrix ρ has elements $\rho_{ij} = \langle c_i^* c_j \rangle$ where $c_{i,j}$ are co-efficients of states $|i\rangle, |j\rangle$ and the angle brackets represent the average of an ensemble

For a two-level system, such as a qubit, $\rho = \begin{pmatrix} \langle |c_1|^2 \rangle & \langle c_1 c_2^* \rangle \\ \langle c_1^* c_2 \rangle & \langle |c_2|^2 \rangle \end{pmatrix}$

The presence of off-diagonal terms in the density matrix is what separates mixed states from pure states. This is central to the concept of superposition.

Fidelity - given two density matrix ρ and σ , the fidelity is define by

$$F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_{tr}$$

Fidelity is a measure of the closeness of two different quantum states. $F \in [0, 1]$ by definition, with 1 being the two states are identical. Thus, in the preparation of entangled systems, a higher F is required.

2.2.2 Physical Realizations of two level systems

A single qubit can be any easily manipulated two-state system. Some familiar examples:

Quantum System	Physical Property	$ 0\rangle$	$ 1\rangle$
Photon	Linear Polarization	Horizontal	Vertical
Photon	Circular Polarization	Left	Right
Nucleus	Spin	Up	Down
Electron	Spin	Up	Down
Two-level atom	Excitation state	Ground state	Excited State

2.2.3 Examples of single qubit quantum gates

Similar to binary logic gates used in classical computation, there exists quantum logic gates which when connected to one another form a quantum circuit. The gates manipulate the superposition of qubits by changing the complex co-efficients of each basis state, resulting in a different state. However, the measurement collapses this superposition (in the Copenhagen interpretation of Quantum Mechanics) returning a set of N classical bits. Here are some common single-qubit quantum gates as examples.

1. The quantum **NOT gate**, represented by **X**, which swaps the amplitude co-efficients of the basis states, represented by the matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
2. The **Z gate** which changes the sign of $|1\rangle$ represented by $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

3. The **Hadamard gate** which turns basis states into superposition states and vice versa. For a single qubit, this is: $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

2.3 Quantum Entanglement

Quantum entanglement is a quantum mechanical phenomena observed in multi-particle systems. If the state of a single particle is dependent on group of particles (thus it cannot be described independently) is said to be entangled. Observations on entangled states are correlated.

Given two different quantum system X and Y (with respective Hilbert spaces H_X and H_Y), the Hilbert Space of the composite system is the tensor product of the Hilbert Spaces $H_X \otimes H_Y$. This system has dimensionality of the product of dimensions of it's constituents i.e. $\dim(H_X \otimes H_Y) = \dim(H_X) \cdot \dim(H_Y)$.

A general state in the composite system $H_X \otimes H_Y$ can be given by

$$|\phi\rangle_{ij} = \sum_{ij} c_{ij} |i\rangle_X \otimes |j\rangle_Y$$

Given states $|\phi\rangle_X$ and $|\phi\rangle_Y$, the composite state is $|\phi\rangle_X \otimes |\phi\rangle_Y$. If states of the composite system can be represent in this form, they are called product states. If the states cannot be represented by a product state, they are called entangled states. Thus, the probabilities of making measurement on either state is correlated.

3 Quantum Teleportation

Quantum teleportation is the communication of information between quantum systems, using entangled quantum systems. Upon creation of an entangled system, which is then physically separated, measurement on a part of a system allows information to be inferred to the other part of the system. This seemingly has no bounds on distance between the system, thus without inspection, may imply possible faster-than-light communication. However, this is not the case. This is encapsulated in the no-communication theorem.

3.1 Bell States

The bell states are four entangled quantum states of two-qubits. These bell states are used as the basis states for measurement. A measurement of one qubit, will assign a value to the other qubit immediately. The four different states are:

$$|\phi\pm\rangle_{XY} = \frac{1}{\sqrt{2}}(|0\rangle_X \otimes |0\rangle_Y \pm |1\rangle_X \otimes |1\rangle_Y)$$

$$|\psi\pm\rangle_{XY} = \frac{1}{\sqrt{2}}(|0\rangle_X \otimes |1\rangle_Y \pm |1\rangle_X \otimes |0\rangle_Y)$$

The four states (00, 01, 10, 11) of the two entangled qubits can be re-written by taking sums of these 4 bell states, with simple algebraic manipulation.

3.2 Teleportation in optical regimes

Teleportation can be used to transfer the quantum state of one photon to another which are physically separated i.e. sending quantum information without a direct exchange of qubits. This is not the same as cloning the photon, which is not possible stated by the no-cloning theorem.

The state a photon expressed as it's polarization as a superposition of two basis states (i.e. $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$) can be communicated from one entity to another through teleportation. A simple method (originally devised by Bennett et al. [3][4],) makes the use of correlated photon pairs emitted by an EPR source. The quantum state a of separate photon can be communicated through making a Bell-State measurement on of one the entangled photons, and the separate photon with arbitrary polarization. This results in one of four possible results, which is communicated classically to the entity with the other entangled photon. By making use of a unitary transformation, this entangled photon can be given the same state as the arbitrary photon $|\psi\rangle$.

4 Decoherence

Quantum computation relies on the ability to manipulate the states, whilst interacting with each other in a isolated setting. The system cannot be totally isolated as that would prevent measurements. Quantum systems are fragile. Upon interacting with the uncontrolled environment, information about the states is lost, as this interaction can be viewed as a non-unitary operation which is not reversible. This interaction collapses the superposition. Thus, precise manipulation of *coherent superposition states* is required, in which there is a definite *phase relation* between the states. The sensitivity of the states with regards to the environment is quantified in terms of decoherence. This is also the reason it is not easy to achieve many entangled qubits.

4.1 Advancements in qubit-entanglement

A way to combat decoherence, is to reduce thermal noise in the environment using superconducting circuits at low temperatures. The lower temperature leads to higher coherence times of the qubits, and thus more time for computation without the risk of decoherence.

10-qubit entanglement has been demonstrated [5], using solid-state architecture. The fidelity of the 10-qubit density matrix is determined using quantum state tomography and found to be 0.668 ± 0.025 . As of 2017, that is the largest entanglement state created using solid-state methods.

5 References and citations

- [1] - Simulating Physics with Computing, Richard P. Feynman
- [2] - Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer, P. W. Shor, (arXiv:quant-ph/9508027)
- [3] Quantum Optics, *An Introduction*, Mark Fox
- [4] Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels, C.H. Bennet et al., Phys. Rev. Lett. 70, 1895 (1993)
- [5] 10-qubit entanglement and parallel logic operations with a superconducting circuit, Chao song et al, DOI: 10.1103/PhysRevLett.119.180511